

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics	COURSE CODE: CSA1501
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I P.Poorna Chandra (192472287)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Poorna Chandra

Assessment Tasks

Application Based Questions

- 1.A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
- 2.An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
- 3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Question 1: A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently.

Relevance to Application

A company that experiences unpredictable traffic spikes during seasonal sales can effectively use cloud auto-scaling and load balancing to manage increased user demand. Auto-scaling automatically increases the number of virtual servers when website traffic grows and reduces them when demand decreases. Load balancing distributes incoming user requests evenly across all available servers, preventing any server from becoming overloaded. This ensures continuous availability, minimizes downtime, and provides customers with a smooth shopping experience during peak sales.

Use of Cloud Concepts

Cloud providers offer services such as Auto Scaling Groups, Load Balancers, and Cloud Monitoring to improve application performance. Monitoring tools continuously monitor CPU usage, memory utilization, response time, and network traffic. When predefined thresholds are reached, additional virtual machines or containers are launched automatically. After traffic decreases, unnecessary resources are removed, reducing cloud infrastructure costs while maintaining application performance.

Practical Insight

A real-world example is Amazon during the Great Indian Festival or Prime Day sale. Millions of customers visit the website at the same time. AWS Auto Scaling automatically launches additional EC2 instances, while Elastic Load Balancer distributes requests across multiple servers. CloudWatch monitors system performance and triggers scaling actions whenever required. This ensures faster page loading, uninterrupted shopping, and successful order processing even during heavy traffic.

Completeness

The combination of auto-scaling, load balancing, and monitoring creates a reliable cloud infrastructure. It improves application availability, maintains consistent performance, reduces operational costs, and increases customer satisfaction. Monitoring tools identify issues early and automatically trigger scaling actions, ensuring business continuity during seasonal sales.

Clarity

Cloud auto-scaling, load balancing, and monitoring work together to create a flexible, scalable, and reliable cloud environment. These technologies help businesses efficiently manage changing workloads while minimizing costs and maximizing customer satisfaction.

Question 2: An organization wants to store large volumes of backup data securely in the cloud.

Relevance to Application

Organizations that store large volumes of backup data require secure, reliable, and scalable cloud storage solutions. Cloud storage services automatically replicate data across multiple data centers to ensure durability and availability. Even if one server or data center fails, backup

data remains accessible from another location. This approach protects important business information from accidental loss, hardware failures, and disasters.

Use of Cloud Concepts

Cloud storage providers use **data redundancy, replication, encryption, and Identity and Access Management (IAM)** to secure backup data. Encryption protects data during storage and transmission, while IAM ensures that only authorized users can access sensitive information. Cloud providers also offer automated backup scheduling and lifecycle management to simplify storage administration.

Practical Insight

A hospital can securely store patient records in cloud storage. If a natural disaster or ransomware attack damages local servers, administrators can restore the data from cloud backups within minutes. This minimizes service disruption and ensures that patient information remains available for medical treatment.

Completeness

Cloud storage provides high durability, scalability, automatic backup, and cost savings compared to traditional storage systems. Organizations pay only for the storage they use, eliminating the need for expensive hardware maintenance while ensuring continuous data protection.

Clarity

Cloud storage is a secure and cost-effective solution for protecting important organizational data. Features such as redundancy, encryption, and access control ensure reliable backup, quick recovery, and long-term data protection.

Question 3: A mobile app company plans to deploy its application globally.

Relevance to Application

A mobile application deployed worldwide requires fast and reliable content delivery. A **Content Delivery Network (CDN)** stores copies of application content on edge servers located near users. This reduces latency, speeds up content delivery, and improves user experience regardless of the user's geographical location.

Use of Cloud Concepts

Cloud providers use CDNs, edge computing, caching, and global data centers to improve application performance. Edge servers process user requests closer to the user, reducing network delays. Cloud providers also use load balancing and automatic failover to maintain service availability.

Practical Insight

Netflix uses a CDN to stream videos worldwide. Content is stored on edge servers located near viewers, allowing videos to load quickly with minimal buffering. Online gaming platforms also use CDNs to reduce lag and improve gameplay.

Completeness

Using CDNs improves application performance, reduces latency, supports global scalability, and ensures continuous service availability. Cloud providers maintain multiple global regions to deliver content efficiently across different countries.

Clarity

CDNs and edge servers make cloud applications faster, more reliable, and easily accessible worldwide. They provide an excellent user experience by reducing delays and improving application responsiveness.

Question 4: A financial institution requires strict compliance and data security in cloud usage.

Relevance to Application

Financial institutions process sensitive customer information and must comply with government regulations. Cloud providers offer secure environments with compliance certifications, auditing, and strong security controls to protect confidential financial data and ensure regulatory compliance.

Use of Cloud Concepts

Cloud providers implement encryption, audit logs, Identity and Access Management (IAM), monitoring systems, and security policies. Logging records every user activity, while encryption protects data from unauthorized access. Organizations can also select cloud regions that meet local data sovereignty laws.

Practical Insight

Banks use encrypted cloud storage for online banking transactions. Every transaction is recorded in audit logs for future verification. Data is stored within approved geographic regions to satisfy government regulations while protecting customer privacy.

Completeness

Cloud security includes encryption, access control, monitoring, compliance reporting, and auditing. Together, these technologies help financial institutions maintain data confidentiality, integrity, availability, and legal compliance.

Clarity

Cloud providers offer comprehensive security solutions that help financial organizations securely process customer information while meeting legal and regulatory requirements.

Question 5: A development team wants to adopt containerization for application deployment.

Relevance to Application

Containerization allows developers to package applications with all required libraries and dependencies into lightweight containers. Unlike virtual machines, containers share the host operating system, making them faster, more efficient, and easier to deploy across different environments.

Use of Cloud Concepts

Container orchestration platforms such as **Kubernetes** automate deployment, scaling, load balancing, health monitoring, and recovery of containers. Kubernetes manages container clusters efficiently and ensures applications remain available even if individual containers fail.

Practical Insight

An e-commerce application can use separate containers for user authentication, product catalog, payment processing, and order management. Kubernetes automatically manages these microservices, ensuring high availability and seamless scaling during periods of high demand.

Completeness

Containers provide faster startup times, lower resource consumption, improved portability, simplified deployment, and better scalability compared to virtual machines. Kubernetes further enhances container management through automation and centralized orchestration.

Clarity

Containerization and Kubernetes provide a modern, flexible, and efficient approach to cloud application deployment. They improve performance, simplify application management, and enable organizations to deploy applications consistently across multiple cloud environments.